A STUDY OF THE PALAEOMAGNETISM OF THE PILANSBERG DYKES

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Summary

A detailed study has been made of the remanent magnetization of five Pilansberg dykes, by means of measurements on oriented specimens of the rocks. Outcrops of the dykes prove to have suffered magnetic disturbance of their original thermo-remanent magnetization. Specimens from depths of a few thousand feet, taken in Witwatersrand gold mines, show highly consistent magnetisation of the basic parts of the dykes over considerable distances between sampling sites. The mean directions given by the five dykes agree well, and the mean direction from the five dykes gives a North-seeking magnetic pole with inclination $+69°{\cdot}3$ and azimuth N 24°·0 E. Assuming thermo-remanent magnetisation by a geocentric dipole field, this places a North magnetic pole in latitude $7\frac{1}{2}°$ N, longitude $42\frac{1}{2}°$ E at the time of intrusion of the dykes. The age of the dykes is uncertain, but is probably about 300 to 400 million years. Some of the specimens have been subjected to alternating magnetic fields in order to test the stability of their magnetisations. The basic specimens are found to be highly stable, being only slightly affected by fields of 100 to 300 oersteds. The possible implications of the results are discussed, in terms of hypotheses of polar wandering and continental drift.

Introduction.—It is well known that igneous rocks frequently possess permanent magnetisation which is a consequence of their having cooled through the Curie points of one or more ferromagnetic constituents in the geomagnetic field. In favourable circumstances such thermo-remanent magnetisation (T.R.M.) may be stable, and the rock will carry a "fossil" trace of the geomagnetic field at the time and place at which the rock cooled. Sedimentary rocks may also exhibit palaeomagnetism, their remanent magnetisation having been acquired through partial orientation of the previously magnetised ferromagnetic grains while settling through still water. Reviews of the extensive work done on palaeomagnetism and rock magnetism generally have been published by Koenigsberger (1938), by Nagata (1953), and by Runcorn (1955).

In 1937 Gelletich published an account of the Pilansberg system of dykes of the Transvaal. He proved that these dykes caused strong and consistent magnetic anomalies of opposite sense to the present vertical field, and used this fact to trace the dykes between outcrops with a Schmidt vertical variometer. Fourteen dykes were identified and traced for varying distances in the Southern Transvaal. They probably form a system radiating to the south and south-east from the Pilansberg, a large body of eruptive rocks constituting the closing phase of the Bushveld igneous activity. A similar system of dykes is known to radiate from

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the Pilansberg to the north-west. Gelletich found that some of the dykes were uniformly basic and of dark colour, while others were composite, having basic rock similar to the simple dykes at both sides, and neutral or slightly acid rock with prominent red or white felspars in a central zone. On transverse traverses the simple dykes showed vertical-field anomalies with one maximum, the composite dykes two maxima. The question whether the composite dykes result from a two-stage intrusion process, as Gelletich believed, or from differentiation *in situ*, is being investigated, as part of a general petrological study of the dykes, by C. B. van Niekerk.* Gelletich identified the ferromagnetic constituent as titanomagnetite. Appropriate studies of the magnetic constituents will be made at a later date.

In the course of nearly 400 traverses Gelletich crossed Pilansberg dykes about 800 times. Only two anomalies were obtained which were not of the characteristic single- or double-peaked form and reversed sense. Two dykes are known to exceed 90 miles in length, being finally lost under increasing thickness of Karroo sedimentary cover to the south, and the parts of the dykes traced by Gelletich are distributed over an area of about 4000 square miles. The existence, as suggested by Gelletich's work, of a consistently magnetised system of rocks over such a large area, affords a problem of great interest. The writer embarked upon the study here reported in 1952, as a result of a suggestion from Professor A. L. Hales.

2. *Method of measurement.*—The magnetometer used for the measurements of remanent magnetisation is of the electromagnetic type, and is modified from a design of the Department of Terrestrial Magnetism, Carnegie Institution of Washington (Graham 1955). The rock specimen is a cylinder of diameter one-half inch and of equal length, with a scratch on the curved surface parallel to the axis. This scratch is an orienting mark to be explained below. The specimen fits in a cylindrical hole in a small cube of perspex, with its scratch in register with a scratch in the cylindrical hole. The cube fits in a cubic hole in the small rotor of a compressed-air turbine of the type described by Beams (1938). By this means the specimen is rotated at 525 revolutions per second close to, and on the axis of, a coil of 27,000 turns. A similar coil, mounted coaxially with the first on the side remote from the specimen, is connected in anti-parallel to eliminate output caused by time-varying spatially uniform electro-magnetic fields. The amplified output of the coils is measured and is proportional to the component of magnetic moment of the specimen normal to the axis of rotation. A sinusoidal pattern is painted in black and white on the cylindrical surface of the rotor, and a streak of light is projected on to this pattern. As the rotor turns, the length of the light-streak reflected from white paint varies sinusoidally. For good wave-form specular reflexion must be avoided. The reflected light is focused on a photocell. The phase difference between the output of the coil and of the photocell is a measure of the angle between some fixed radius of the rotor and the component of magnetic moment normal to the axis of rotation. The output of the photocell is passed through a phase-shifting circuit to the X plates of a cathode ray oscilloscope. The amplified coil output is applied to the Y plates, and measurement of an angle consists in setting the phase shift of the photocell output so as to reduce the resulting ellipse to a straight line. Repetitions are easily made to better than one degree, and the uncertainty in orientation of the specimen in the cube is the principal source of measurement error.

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A small standard magnet mounted in a graduated perspex cylinder replaces the specimen for calibration of the phase-shifting controls in terms of angle. This magnet is periodically standardised at the Magnetic Observatory, Hermanus, through the kindness of Mr A. M. van Wijk, Officer in Charge, and so serves to calibrate the magnetometer for intensity as well as for angle.

The photocell output is also taken to a frequency discriminator which enables the compressed air supply to be adjusted so as to keep the angular velocity of the rotor to 525 ± 2 r.p.s. while the reading is taken. The use of inductances is avoided in the phase-shifting and frequency discriminating circuits, so as to minimise cross-feed to the coils.

A tuned stage set to 525 c.p.s. is included in the amplifier, in order to improve the signal to noise ratio, and the photo-electric signal is passed through a similar tuned stage so as to reduce to a low value the rate of change of phase difference with frequency. The requirement of matching of the two tuned stages limits the sharpness of tuning which it is convenient to use. With the existing amplifier, the lower limit of intensity which may be measured is about 10^{-5} gauss. This is more than sufficient for these dykes, but could be improved by improving the amplifier.

For each specimen, the measurement procedure is carried out with the cube in six different positions within the rotor. Three pairs of angles are so obtained, the sum of each pair of which must be 360°. Small corrections are applied to make the sum of each pair 360°. This corrects the readings for temperature drift, and amounts to continuous calibration of the magnetometer. The constancy of the corrections provides a check of the accuracy of settings. Measurement of a specimen takes ten to fifteen minutes. One angle from each pair is used in plotting the direction of magnetisation. The three angles give, in the stereographic projection, a straight line and two arcs which should intersect in a point. In nearly all cases the intersection was good, but occasionally there were discrepancies of up to about ten degrees. It is thought that such discrepancies arise where the distribution of intensity within the specimen is asymmetrical.

The electromagnetic type of magnetometer is convenient for use in buildings, and in cities, where large varying magnetic fields cannot be avoided. An improved magnetometer of generally similar type to that described has been constructed by Professor A. L. Hales, Mr J. A. Keiller, and Mr K. W. Graham of the Bernard Price Institute, and is capable of measurements down to intensities of 3×10^{-7} gauss, despite the bad magnetic disturbances characteristic of the site.

The specimens were secured by drilling with a diamond coring crown giving a core of diameter one-half inch. The surface outcrop specimens were drilled with vertical axes and were oriented in azimuth before removal by means of a diamond scratch on the curved surface, the radius through the scratch being made the sun's azimuth by means of a simple sun-dial instrument. The underground specimens were drilled with horizontal axes and an orienting instrument was fitted on each core. By means of a plumb-line on this instrument, a diamond scratch was scribed on the curved surface at the end of the downward radius. A graduated dial on the orienting instrument enabled readings to be taken with a suitably placed theodolite, which gave the azimuth of the axis of the specimen with reference to the mine survey network. After removal, specimens were cut to length with a suitable diamond saw. The magnetometer measurements were carried out in the writer's laboratory in the Bernard Price Institute of Geophysical Research.

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3. *Work on surface outcrops.*—About 200 specimens were taken during 1952–3 from ten surface outcrops of five of the dykes listed by Gelletich. These specimens showed large, apparently random scatter in both direction and intensity of remanent magnetisation. The directions were distributed over the whole sphere. Intensities covered the range from 1×10^{-3} to 570×10^{-3} gauss. Reference to Fig. 7 will show that the maximum intensity encountered in any dyke underground was 14×10^{-3} gauss. It is clear, therefore, that the superficial parts of the outcrops have been remagnetised by some agency and no longer preserve any T.R.M. they may have had. The incidence of lightning flashes to ground is high in the Transvaal, and lightning may play a part in remagnetising surface rocks. Remagnetisation may also have occurred during oxidation of the magnetic constituents, though in this case there appears to be no obvious explanation for the random directions.

It was evident that the random directions observed could not be typical of the body of the dyke, in view of the consistent anomalies found by Gelletich, and advantage was therefore taken of the opportunity provided by the gold mines of the Witwatersrand to sample some of the dykes at depth.

4. *Measurements on underground specimens.*—Sets of oriented specimens have been secured from six gold mines traversed by Pilansberg dykes. Specimens were taken across the whole width of the dyke at each level, and wherever possible several levels were sampled in order to make the data representative of the largest possible volume of rock. In some cases, however, the mine authorities could provide facilities at one or two levels only.

The dykes are best considered separately. The data have been combined using the statistical treatment of dispersion on a sphere given by Fisher (1953). It is assumed that the error distribution is such that the frequency density is proportional to $e^{\kappa \cos\theta}$, where θ is the angle between the population mean direction and an observed direction, and κ is a measure of precision. If R is the vector sum of N unit vectors distributed in this way, there is a probability P that the resultant of the N vectors will depart from the population mean direction by an angle exceeding α where

$$1-\cos\alpha = \frac{N-R}{R}\left[\left(\frac{1}{P}\right)^{\frac{1}{N-1}} - 1\right]$$

P is taken as 0·01 in this paper, so that the values of α may be described as radii of circles of 99 per cent confidence for the mean directions.

The parameter κ is useful in comparing sets of data. An estimate of κ is given by

$$k=(N-1)/(N-R).$$

The data were plotted graphically. To find the direction cosines, the inclination and azimuth for each specimen were read off the plot. Errors introduced in this procedure are small compared with the errors discussed in Section 5 below.

Wherever possible specimens were taken in pairs one or two inches apart. This was done so that any variation in direction which might occur over a few feet, such as might have been caused by flow below the Curie point, could be distinguished from variation brought about by other causes. In fact no evidence was found of any such effect. As a check on the validity of including both specimens of each pair in the statistical treatment, the data of Dykes A and C have been combined using firstly both, and secondly one, of each pair of specimens. The

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results indicate that the specimens of a pair may be treated as independent, and all of the specimens have been used in obtaining the results given below.

The figures given for the widths of dykes refer to the distance between contacts in the drive, and take no account of the angle of intersection. In some cases the width of the dyke will therefore be less than the figure given.

Dyke A. Libanon Gold Mine

This dyke is composite, having basic outer zones and a red felspathic centre. In discussing this and the other dykes, the zones will be distinguished by the terms "basic" and "felspathic", in order to avoid any attempt at this stage to classify the rocks as syenite, diorite, dolerite, or any other petrological type. The dyke is either Gelletich's Venterspost Dyke or is connected to it. It has been described by Lombaard (1939). Specimens were taken from two levels, at depths of 2704 and 3929 feet, the slant distance between the two being 7025 feet. The width of the

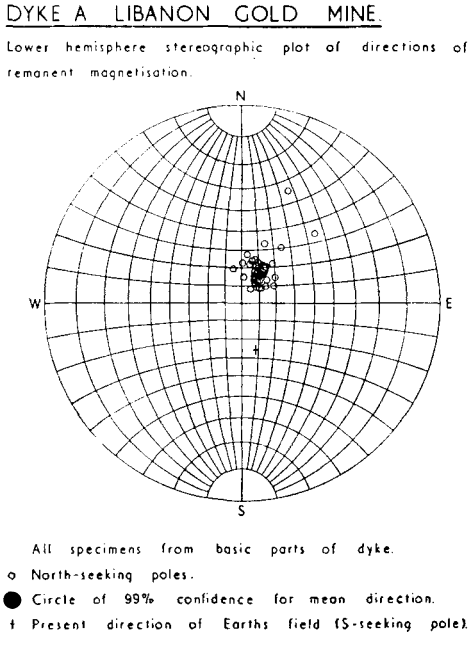


Fig. 1.

dyke was 176 feet at the upper level and 116 feet at the lower, the felspathic zone being 2 feet and 21 feet wide respectively. The felspathic specimens proved to be too weakly magnetised for measurement, though in some cases there was a detectable moment. Thirty-six basic specimens have been measured, and the directions of remanent magnetisation are shown in Fig. 1. In all cases the north-seeking poles are directed downwards. So far as the vertical field component is concerned, this agrees with Gelletich's anomalies in opposition to the present field.

The specimens from the two levels have been separately combined, using Fisher's statistics. Since κ is large, the distribution approaches the normal distribution, and it has been assumed that a Student's t-test may be applied to test the significance of the difference between the mean directions for the two levels. When this is done, the difference of $7°{\cdot}7$ proves not to be significant at the one

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per cent level. Combination of the 36 specimens from the dyke gives the data shown in Table I below. The distribution of intensity of remanent magnetisation is shown in Fig. 7.

Dyke B. Venterspost Gold Mine

This is again a composite dyke, and is probably connected with Dyke A, the two mines being adjacent. The contact between basic and felspathic zones is very sharp. Specimens were taken from two levels, at depths 2259 and 2602 feet, the slant distance between them being 700 feet. The dyke is relatively narrow; the widths at the two levels were 35 feet and 59 feet respectively, the felspathic zone occupying about half the total width. In this, as in all of the dykes, the number of specimens was dictated by the difficulty of finding sound patches in the generally fractured walls of the drive, and only 13 felspathic and 8 basic specimens were secured. The felspathic specimens were weakly magnetised, and only five

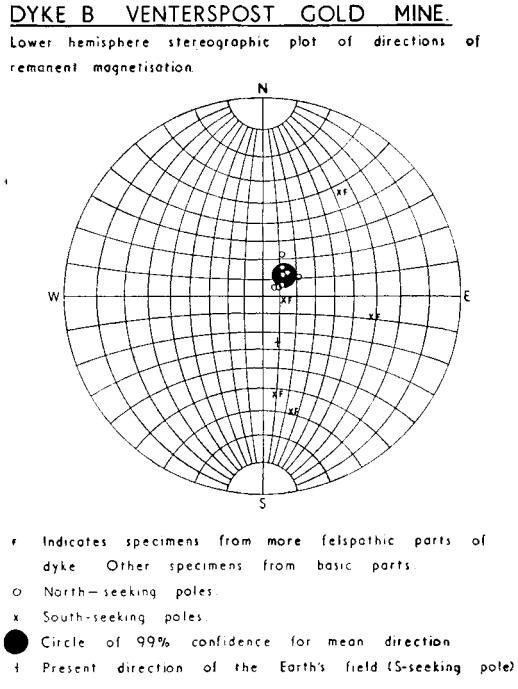


Fig. 2.

of them could be measured. The intensities for these five range from 0·012 to 0·166 × 10^{-3} gauss; Fig. 7 gives the intensities of the basic specimens, for comparison. Fig. 2 gives the directions of magnetisation of these five felspathic specimens and of the eight basic specimens. The felspathic specimens have south-seeking poles widely scattered in the lower hemisphere of projection, whereas the basic specimens give north-seeking poles downward, in a close group. It will be shown in Section 6 below that there is reason to suspect that the felspathic specimens are magnetically unstable, and in fact one felspathic specimen from this dyke was the only specimen from any dyke whose direction of magnetisation is known to have altered while it was stored in the laboratory, the change being 15 degrees. Because of their suspected instability, the felspathic specimens are not considered reliable

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for palaeomagnetic purposes, and are disregarded. The basic specimens from the two levels gave mean directions which did not differ significantly. The eight basic specimens from this dyke give the mean direction and measures of dispersion given in Table I, and the intensities of magnetisation shown in Fig. 7.

Dyke C. Robinson Deep Gold Mine

Dyke C is the well-known Robinson Dyke. Where it is exposed in the mine it is generally of basic rock, though there is a general increase of felspars towards the centre. The felspars are white and pink. Forty-nine specimens were taken from four levels, at depths ranging from 5700 to 8300 feet. The slant distance between the highest and lowest levels was about 5200 feet. The width of the dyke varied between 241 and 304 feet. Only on one level was rock encountered which showed a sufficient change in composition from the rest of the dyke to be described as felspathic. Seven specimens of this felspathic region gave intensities of remanent magnetisation ranging from 0·018 to 0·038 $\times 10^{-3}$ gauss, and showed wide scatter in direction. Reference to Fig. 7 will show that these specimens differ in intensity

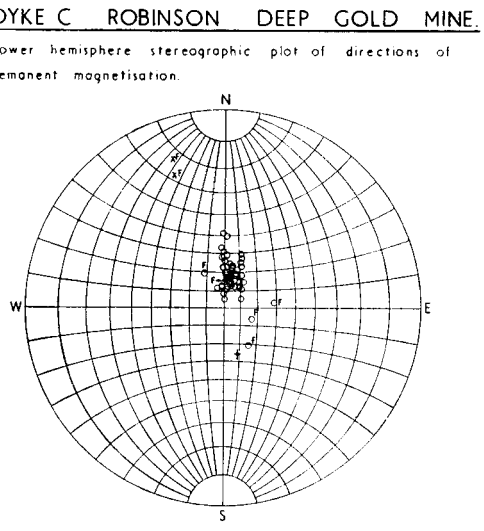


Fig. 3.

by a factor of about 100 from those from the basic rock. This fact together with the wide scatter in directions makes it appear reasonable to regard the felspathic rock as differing in magnetic properties from the rest of the dyke, and to exclude it from consideration.

The observations on basic specimens from each level were combined to give mean directions and values of α, and a Student's t-test was applied, as in the cases of Dykes A and B, to test for the significance of the differences between the mean directions. The difference of 14°·0 between the uppermost and the lowest sites

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proved to be significant at the one per cent level. The other five differences were not significant. It is not very surprising to find a significant difference in direction of magnetisation between two places nearly a mile apart, for the dates of passage through the Curie temperature may well have differed by a length of time sufficient for secular variation to produce a change in field direction of a few degrees.

The 42 basic specimens are all magnetised with north-seeking poles downward and towards the North. The directions of magnetisation are shown in Fig. 3 and the intensities in Fig. 7. The mean direction and measures of dispersion for the basic specimens from the four levels taken together are given in Table I.

Dyke D. Simmer and Jack Gold Mine

This large composite dyke is known on the mine as the Syenite Dyke. The name "Goch Dyke", which belongs to another dyke in the same mine, was previously applied to Dyke D in error by the writer, and it is mentioned as such by Runcorn(1955). Dyke D is Gelletich's Number 8. It has basic outer zones and

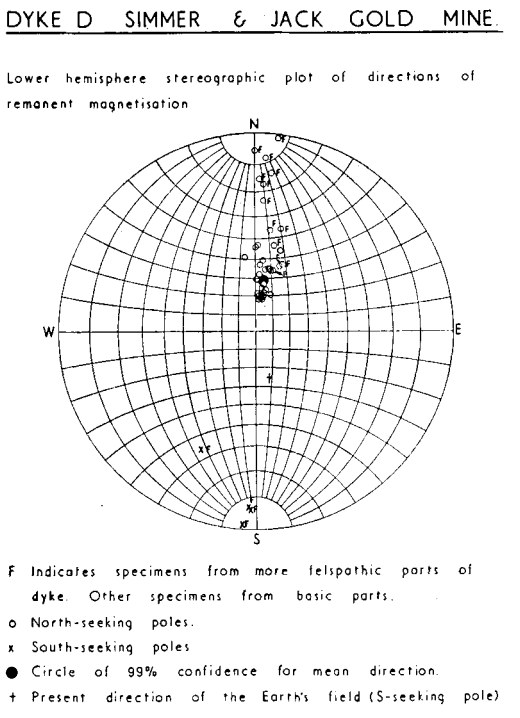


Fig. 4.

a central zone with prominent red felspars. Near ground level the inner zone is nearly pure felspar, and the contacts with basic zones are sharp. At greater depths the central rock is intermediate in composition between the pure felspar and the basic rock, and the transitions are less abrupt. For reasons of electrical power supply it was possible to drill specimens only on one level, at a depth of 6037 feet. The dyke was there 201 feet wide, and the felspathic zone 50 feet. Forty-two specimens were secured, of which 17 were felspathic and 25 basic. The felspathic

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specimens gave intensities ranging from 0·15 to 0·85 × 10^{-3} gauss, so that their intensities are weaker than those in the basic rock by a factor of 10 (see Fig. 7).

The directions of remanent magnetisation are plotted in Fig. 4. The basic specimens give the usual close group of north-seeking poles directed downward in the north-east quadrant. The felspathic specimens show directions having a remarkable distribution close to the plane of the meridian. This distribution would appear in the sum of two vectors directed, respectively, in the direction shown by the basic specimens, and approximately in the direction of the present geomagnetic field, if the ratio of the lengths of the vectors varied. The observations can therefore be accounted for by supposing that the felspathic rock contains two ferromagnetic constituents, one bearing stable T.R.M. in the same direction as the basic specimens, and the other contributing a magnetisation in the direction of the present field. The second component could be due either to isothermal remanent magnetisation (I.R.M.) or to some process involving oxidation or other chemical change. The phenomenon would then resemble that found by Kawai and Kume and reported by Nagata (1953, p. 210).*

The alternative possibility exists that the data record the geomagnetic field in the process of a reversal which took place rapidly enough for large changes to take place during the cooling of the dyke. It was hoped that the demagnetising experiments described in Section 6 below would show either that the felspathic rock was as stable as the basic, or that it had a very unstable component. Unfortunately the results were inconclusive, indicating only slightly lower stability for the felspathic rock of Dyke D. There seems to be no evidence of any really "soft" magnetic constituent likely to acquire I.R.M. However it remains possible that the rock has in some way acquired remanent magnetisation in recent times, and the meaning of the data must remain in doubt.

The directions given by the 25 basic specimens have been combined in the usual way. The direction of the mean and its confidence limit are given in Table I. The intensities of magnetisation of the basic specimens are shown in Fig. 7.

Dyke E. East Geduld Gold Mine

This is Gelletich's Dyke 10. In the East Geduld Mine it is entirely basic, and at the date of sampling it was the opinion of the staff geologists of Union Corporation that it might be a dolerite dyke of Karroo age. However, Professor F. Walker of the University of Cape Town very kindly examined some thin sections for us, and gives the following respects in which the rock differs from typical Karroo dolerite:—

(*a*) Grain-size coarser;
(*b*) Texture less ophitic;
(*c*) Considerably more modal iron ore;
(*d*) Much higher degree of alteration, probably hydrothermal, to amphibole, chlorite, and epidote.

Professor Walker therefore concludes that the petrology is consistent with the view that the dyke is of late Bushveld Igneous Complex age, and does not support the view that it is of Karroo age.

* The writer is indebted to Professor P. M. S. Blackett, F.R.S., for drawing his attention to this resemblance.

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The dyke has been examined, and hand samples taken, in the adjoining Modderfontein East Mine, and it there contains large volumes of the red felspathic rock which is characteristic of Dykes A, B, C and D. Mr C. B. van Niekerk and Dr L. O. Nicolaysen, who kindly accompanied Mr van Niekerk to Modderfontein East Mine, concluded that the dyke was definitely not Karroo dolerite. They had reached the same conclusion as Professor Walker on a preliminary study of thin sections. Magnetometer work by Weiss on East Geduld and adjoining properties confirms Gelletich's consistent anomaly opposed to the present field, whereas a larger dyke of agreed Karroo age (Gelletich's Dyke D) gives a consistent anomaly in the sense of the vertical field. The dykes intersect in Modderfontein East Mine, and it is most unfortunate that the workings do not expose any contact between them sufficiently for the establishment of a relative age on the basis of cooled zones. However the combined evidence from petrology and surface magnetometry appears conclusive that the dyke is a member of the Pilansberg system.

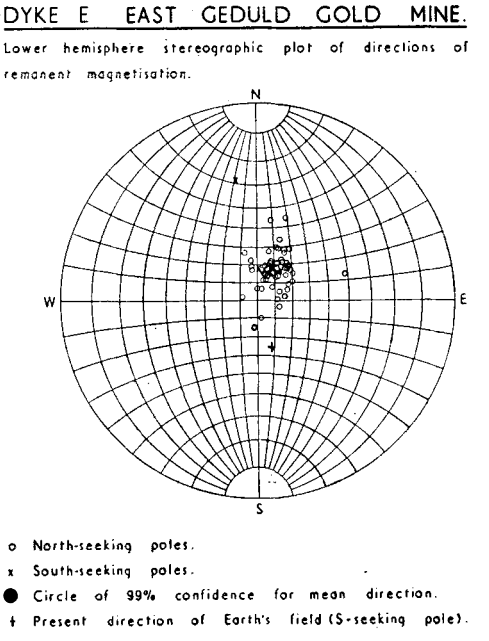


FIG. 5.

Sixty oriented specimens were secured from four levels in the East Geduld Mine, the depths being 2973, 3277, 3473 and 3677 feet. The slant distance between the highest and lowest sampling sites was 6400 feet. At the deepest level the dyke has two branches, 86 and 104 feet wide respectively, separated by 54 feet of country rock. On the other three levels the width was 117, 172 and 173 feet respectively.

Two of the differences between the mean directions of magnetisation for the separate levels prove to be significant at the one per cent level, the differences being 9·3 and 11·3 degrees. The other four differences are not significant. The directions of magnetisation given by the 60 specimens are shown in Fig. 5. Fifty-eight give north-seeking poles downwards, and two south-seeking poles. The

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specimen showing a south-seeking pole in the group of north-seeking poles is the only oriented specimen from this dyke having traces of red felspar in it, though it cannot be described as markedly felspathic. As remarked above, in the adjoining mine the dyke has large felspathic regions. The other specimen giving a south-seeking pole in the lower hemisphere appears to be of normal basic composition. These two specimens have been excluded from the combination of directions, on the ground of their wide divergence from the group formed by the remainder. Two specimens from a quartzite xenolith also showed south-seeking poles downward; these are not included in the figure. The mean direction and values of k and α for 58 specimens are given in Table I, and the intensity distribution in Fig. 7.

Dyke F. Blyvooruitzicht Gold Mine

Over 100 oriented specimens were secured from a dyke in this mine which is believed to be the Oberholzer Dyke. The directions of magnetisation were widely scattered and the demagnetisation tests described in Section 6 below showed that the magnetisation was generally soft, though possibly there are hard and soft components of magnetisation.

The Remanent Magnetisations of Five Pilansberg Dykes

Table I gives the results of the statistical combinations of the directions of magnetisation in the basic rock of each of five Pilansberg dykes. In Fig. 6 the circles of confidence for the mean directions given by the five dykes are shown. It will be seen that the agreement is close, and supports the belief that the dykes

Table I

Mean Directions of Remanent Magnetisation in Basic Parts of Five Pilansberg Dykes

Dyke	S Lat. °	′	E Long. °	′	N	I deg.	A deg.	α deg.	k
A	26	21	27	36	36	68·2	30·2	4·6	44·2
B	26	17	27	39	8	73·1	45·5	7·0	110·1
C	26	13	28	03	42	71·8	14·1	4·9	33·0
D	26	12	28	08	25	60·6	7·9	3·3	122·7
E	26	13	28	27	58	70·1	31·4	3·8	38·4

The latitude and longitude refer to the place of origin of the specimens.
N = number of specimens
I = inclination of north-seeking pole of mean direction
A = azimuth of north-seeking pole of mean direction from North through East
α = radius of circle of 99 per cent confidence for the mean direction
$k = (N-1)/(N-R)$.

form a geologically related system and are of the same age. It should be noted that the five mines from which samples were taken are distributed approximately along an east-west line some 54 miles long. Over this distance the magnetising field was clearly closely uniform in direction.

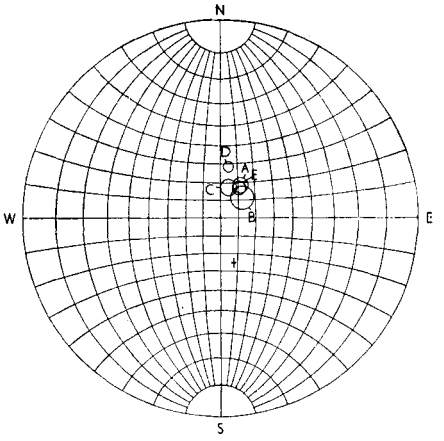
The mean direction resulting from combination with equal weights of the five mean directions has inclination $+69°{\cdot}3$ and azimuth $24°{\cdot}0$ east of north for the north-seeking pole. If the dykes acquired T.R.M. in a geocentric dipole field,

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the North magnetic pole at the date of intrusion was in latitude $7\frac{1}{2}^\circ$ N, longitude $42\frac{1}{2}^\circ$ E, that is in eastern Abyssinia.

MEAN DIRECTIONS OF REMANENT MAGNETISATION OF FIVE PILANSBERG DYKES

Lower hemisphere stereographic plot.



○ Circle of 99% confidence for the mean direction in the basic parts of the dyke indicated. North-seeking poles downward in all cases.

+ Present direction of the Earth's field (South-seeking pole).

FIG. 6.

5. *Errors.*—The variation in the observed directions may be ascribed to four sources:—

1. Variation in the direction of magnetisation in the rock.
2. Changes in the direction of magnetisation caused by the drilling procedure.
3. Error in the orientation procedure in the mine.
4. Measurement error.

Only the fourth was easy to study systematically. Measurement error includes error in placing the specimen in the perspex cube, error in setting for no phase difference, variation in the magnetometer and error in the graphical solution. All of these operations were repeated for each of 6 specimens from Dyke C, 5 specimens from Dyke D and 7 specimens from Dyke E. The mean direction and value of α were calculated for each group as given by the first and as given by the second set of readings. Table II shows the differences, Δ, between the mean directions given by the two sets of readings, and the values of α. It is clear that the measurement error is negligible compared with the other sources of variation.

TABLE II

Differences on Repetition

Dyke	N	Δ	α_1	α_2
C	6	0° 00′	8°·5	8°·9
D	5	0° 00′	10°·6	10°·1
E	7	1° 59′	7°·3	8°·8

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Rather elaborate experiments would be necessary in order to measure the error contributed by the orientation procedure. The orientation error will arise mainly from (*a*) departure of the core from straightness, caused by distortion of the drill frame, (*b*) departure of the axis of the core from the horizontal. It is estimated that these effects will cause an orientation error of four degrees for one specimen in a hundred. If this estimate is correct, it is clear that most of the observed variation

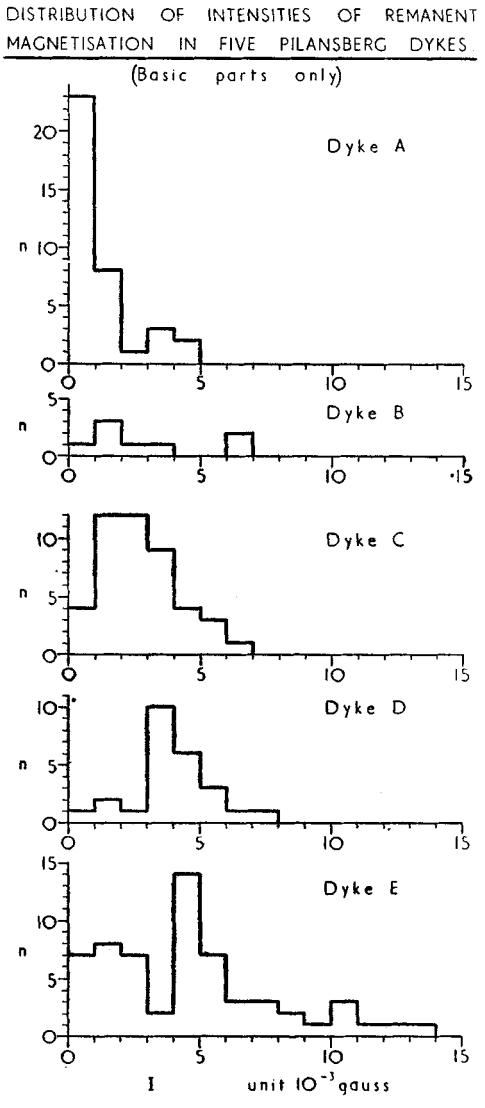


FIG. 7.

arises from sources (1) and (2). The drilling process involves no large mechanical shocks, and may probably be disregarded, as the variation in the rock includes any effect of the blasting incidental to tunnelling through the dyke. It is therefore reasonable to conclude that most of the observed variation is true variation in the direction of magnetisation in the rock, though no estimate can be made of the contribution of blasting to this variation.

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6. *Stability of the remanent magnetisation.*—The best evidence of the stability of the remanent magnetisation of the basic parts of the Pilansberg dykes is their magnetisation in closely grouped directions making angles of about 140 degrees with the present geomagnetic field. As an additional test, a number of specimens from each dyke were subjected to alternating fields in order to find out what field would cause given changes in the magnetisation of the specimen. The demagnetising field was produced by passing 50 c.p.s. current through a coil enclosing the specimen. The specimen was first measured in the magnetometer, and then subjected to an alternating field which was slowly reduced from 10 oersteds to zero, the geomagnetic field being nullified, after which the magnetometer measurement was repeated. The procedure was then repeated with initial peak field 20, 50, 100, 200 and 300 oersteds. Only 31 specimens were used in these tests, because the remainder are being reserved for work on the thermal demagnetisation and remagnetisation characteristics. For this reason no study could be attempted of the effect of variation of the angle between direction of remanent magnetisation and direction of the applied field. This angle was however varied without any marked effect.

TABLE III

Effects of Alternating Magnetic Fields on Remanent Magnetisation

The table shows the number of specimens

(P) whose directions of magnetisation are changed by 10° by an alternating magnetic field reduced slowly from an initial peak value in the range shown;

(Q) whose intensities of magnetisation are reduced by 20 per cent by an alternating magnetic field reduced slowly from an initial peak value in the range shown.

Dyke	Rock type		Initial peak field, oersteds						
			0–10	10–20	20–50	50–100	100–200	200–300	>300
A	Basic	P					4	1	
		Q					5		
B	Basic	P						1	1
		Q				1	1		
	Felsp	P	1				1		
		Q			2				
C	Basic	P					1	2	4
		Q					2	3	2
D	Basic	P				1	1	2	1
		Q					1	1	3
	Felsp	P			1	1	2	1	
		Q				2	2	1	
E	Basic	P							5
		Q				1		1	3

Table III gives the results of these demagnetising experiments. It will be seen that:—

(1) The basic specimens require fields of the order of hundreds of oersteds to disturb their remanent magnetisations to any considerable extent. This gives

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strong support to the view that the remanent magnetisation of the basic rock is T.R.M. (cf. Nagata 1953, p. 140).

(2) The felspathic specimens are affected by smaller fields than the basic specimens from the same dyke, though the distributions overlap and some felspathic specimens from Dyke D show high stability. This last fact is in accord with the fact that the directions for the felspathic specimens include some within the group formed by the basic specimens (see Fig. 4). Similarly, Fig. 2 shows that in Dyke B all the felspathic specimens give widely different directions from the basic specimens, and Table III shows a large difference in stability. The demagnetising tests therefore support the exclusion of the felspathic specimens as being of suspect stability.

(3) There is a general tendency for direction of magnetisation to be less disturbed than intensity by a given field.

7. *Discussion.*—As a result of Gelletich's work, the Pilansberg dykes have many times been referred to in the literature as an instance of reversal. This has been shown to be incorrect. The situation revealed by the present work is quite different from, and more surprising than, reversal. It is amusing to speculate on Gelletich's reactions had he ever made a traverse with a horizontal component magnetometer and found an anomaly of normal sign. The essential point is that the magnetic vector dips down towards the north. In view of the high consistency of the results, and of the high stability with regard to A.C. fields, it is reasonable to regard the remanent magnetisation as T.R.M. at least provisionally. It is intended to study the magnetic behaviour of the rock as a function of temperature, and the usual X-ray and other analyses will be carried out. Whatever the outcome of these investigations may be, there is so far as is known no mechanism known by which a body of rock distributed over an area of thousands of square miles may become magnetised in a highly constant direction other than the direction of the Earth's magnetic field. The mechanisms proposed by Néel (1951, 1955) and by Graham (1953), as well as that found in a piece of dacitic pumice by Nagata (1952, 1953), provide only for reversal of sense; and even if, to suppose the improbable, the whole mass of the Pilansberg dykes satisfies some one of Néel's conditions, we still have to account for a South magnetic pole in the region of Abyssinia. The only escape from the conclusion that a magnetic pole was in this region appears to lie in supposing that the field at the time of intrusion was not dipolar at all.

The question whether these dykes exhibit normal T.R.M., or acquired magnetisation in reverse sense to the magnetising field, is therefore less important than in cases of magnetisation in reverse to the present field. However, it will be of great interest to find out how rare or how general may be such magnetite complexes as that studied by Nagata, and for completeness it is desirable to try to establish the sign of the magnetising field. The usual heating experiments will therefore be done.

In order to place the results here reported in their context, it may be desirable briefly to summarise the present state of the development of the palaeomagnetic investigation. There is general agreement from the work on glacial varves that since the last Ice Age the geomagnetic field direction has not changed very markedly from its present direction, apart from changes in inclination which may be a consequence of the process of deposition, and changes in declination which indicate that the field direction varies with an average close to the direction of an axial geocentric dipole field. In the Tertiary and early Quaternary we have the evidence

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collected from lavas in Iceland by Hospers (1951, 1953, 1954), in France by Roche (1950, 1951), in Ireland by Hospers and Charlesworth (1954), and in Oregon and Washington by Campbell and Runcorn (Runcorn 1955), for a number of reversals of the geomagnetic field, the lava flows showing directions of magnetisation grouped round the dipole field direction, with groups of flows bearing alternately the two senses of magnetisation. It remains possible that the reversals arise from special properties of the ferromagnetics, and not from reversals of the field. However, Hospers has shown that the Iceland lavas show no similarity in their behaviour on heating to Nagata's dacite, and it appears that the evidence of reversals of the Earth's field is accumulating steadily. The observations of Torreson, Murphy and Graham (1949) on Tertiary sediments contradict this, showing generally a dipole field of the present sign. Where direction and sense of magnetisation agree approximately with the present field, proof of stability is of special importance. Since Torreson, Murphy and Graham are not able to produce such evidence, their work cannot be held to disprove reversals. The lavas carry their own evidence of stability, in the persistence of the reverse magnetisation.

Palaeomagnetic data from older rocks are much fewer, but are sufficient to give strong indication of a further result of palaeomagnetic enquiry of even greater interest than the geomagnetic reversals. Irving's (1954, reported also in Runcorn 1955) remarkable study of the late Pre-Cambrian Torridonian sandstones of Scotland is of great importance. Irving found mean directions of magnetisation in both senses dipping downwards to the south-east. The stability of the remanent magnetisations is attested both by the large angles the direction makes with the present field, and also by Irving's application of both the conglomerate test and the fold test proposed by Graham (1949). On the assumption of a geocentric dipole field these data indicate a magnetic pole, of alternating sign, in the region of Abyssinia. Creer's (1955, reported also in Runcorn 1955) data for the Old Red Sandstone of South Wales, which is of Palaeozoic age, indicate on the dipole assumption a North magnetic pole in the South Atlantic. Clegg, Almond and Stubbs (1954; see also Runcorn 1955) find that Triassic sediments in England indicate magnetic poles of both signs in the South Atlantic, and Graham (1955) reports that Permian and Triassic sediments in the south-western United States indicate, on the assumption of a dipole field, a North magnetic pole in the same region as given by Clegg, Almond and Stubbs.

The age of the Pilansberg dykes is unfortunately uncertain. They are certainly pre-Karroo, and according to du Toit (1954) they are post-Waterberg. According to Cooke (1953) this places them between 300 and 400 million years in age, which would make them Palaeozoic. However, the area where they are exposed has been subject to erosion for a long and uncertain period, and until good radioactive age determinations become available their age must be in doubt. The possibility cannot be excluded that they are of late Pre-Cambrian age, so that the agreement in the derived position of the pole between Irving's data and those here reported is of some interest.

The data from the older rocks suggest strongly that there must have been considerable relative movements between the axis of the geomagnetic field and the continents. If it be assumed that the magnetic poles do not depart by large angles from the axis of rotation, the data suggest either polar wandering with the continents remaining fixed relative to one another, or continental drift, or a combination of the two. Creer, Irving and Runcorn (1955; see also Runcorn 1955) have suggested

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that the existing data can be accounted for by polar wandering. Their tentative locus for the poles starts in late Pre-Cambrian time with the South geographic pole, located from Irving's data, near the position given by the Pilansberg dykes for the North magnetic pole. If it be allowed that geomagnetic reversals do occur, and allowing for the uncertainty in dating, the present results may therefore fit the polar locus suggested.

There exists the alternative possibility of continental drift, which is being studied in particular by Professor P. M. S. Blackett. When sufficient data become known from various continents it should become possible to discriminate, but in the present state of knowledge both hypotheses remain open, so far as the palaeomagnetic data are concerned.

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